



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Guwahati

Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH

**POWER ELECTRONICS AND INSTRUMENTATION
ENGINEERING**

3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure (From Academic Session 2024-25 onwards)

B. Tech 3rd Semester: Power Electronics and Instrumentation Engineering

Semester III

Sl. No.	Course Code	Course	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	PE241301	Semiconductor Devices and Circuits	3	0	0	3	30	70
2	PE241302	Digital Electronics	3	0	0	3	30	70
3	PE241303	Electrical and Electronics Measurements	3	0	0	3	30	70
4	PE241304	Network Analysis	3	1	0	4	30	70
5	MA241305B	Mathematics III-B	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	-	50
7	AC241307	Language other than MIL	1	1	0	0 PP/NP	-	50
Practical								
1	PE241311	Semiconductor Devices and Circuits Lab	0	0	2	1	15	35
2	PE241312	Digital Electronics Lab	0	0	2	1	15	35
3	PE241313	Electrical and Electronics Measurements Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP (Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241301	Semiconductor Devices and Circuits	3-0-0	3

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Explain the behaviour of semiconductor devices
- **CO2:** Analyze the functioning of various solid-state devices, including diodes, BJTs and FETs.
- **CO3:** Illustrate the biasing techniques for BJTs and FETs
- **CO4:** Realize simple amplifier circuits using BJTs and FETs.
- **CO5:** Explain the behaviour of special semiconductor devices

MODULE 1: Overview of Semiconductor Physics: (6 Lectures)

Review of Quantum Mechanics, Electrons in periodic Lattices, E-k diagrams; Energy bands in intrinsic and extrinsic silicon; Carrier transport: diffusion current, drift current, mobility and resistivity; sheet resistance, design of resistors; Generation and recombination of carriers; Poisson and continuity equation.

MODULE 2: P-N junction: (10 Lectures)

The open circuited junction, space charge region, the biased p-n junction, I-V characteristics, Piece-wise linear characteristics & equivalent circuits, Diode resistance, Capacitance, switching time, small signal switching models; Avalanche breakdown, Zener breakdown. Half wave and Full wave single phase rectifiers, filtering and their analysis, peak inverse voltage, voltage multiplier circuits, Clipping and Clamping circuits. Zener diode and voltage regulator, Schottky diode, Varactor diode

MODULE 3: Bipolar Junction Transistor: (12 Lectures)

PNP and NPN junction transistors, various configurations of BJT and their input & output characteristics, different modes of operation, Ebers-Moll representation, Early effect, Avalanche breakdown & Punch through. BJT biasing: Operating Point, DC & AC load lines, analysis of different biasing circuits and problems, Stabilization, Thermal runaway and thermal stability, BJT as a switch and amplifier.

MODULE 4: Field Effect Transistor: (8 Lectures)

Structure and operation of JFET, the drain and transfer characteristics, Biasing of FETs. MOSFET: Structures and operations of Depletion and Enhancement types of MOSFET, the drain and transfer characteristics, MOS capacitor, C-V characteristics, introduction to CMOS.

MODULE 5: Special Semiconductor Devices: (4 Lectures)

An Introduction to Photonic devices: LED, LASER diodes, Photo detectors, Solar Cells. An Introduction to Power Semiconductor Devices: Power BJT, Power MOSFET, UJT, CCD, Thyristor, Tunnel diode.

Suggested Text/ Reference Books:

1. G. Streetman, and S. K. Banerjee, "Solid State Electronic Devices," 7th edition, Pearson, 2014.
2. D. Neamen, D. Biswas, "Semiconductor Physics and Devices," McGraw-Hill Education
3. S. M. Sze and K. N. Kwok, "Physics of Semiconductor Devices," 3rd edition, John Wiley & Sons, 2006.
4. A.S. Sedra and K.C. Smith, Microelectronic Circuits, Saunder's College Publishing, 1991.

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241302	Digital Electronics	3-0-0	3

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Apply the Boolean algebra and K map to minimize logic function.
- **CO2:** Design and analyze combinational logic circuits.
- **CO3:** Design & analyze sequential logic circuits.
- **CO4:** Explain the logic families and memory devices

MODULE 1: Introduction to Digital Circuits (5 Lectures)

Data and number system: Binary, Octal and Hexadecimal representations and their conversion, BCD, ASCII, EBCDIC, Gray codes, code conversion, Representation of Signed binary numbers with 1's and 2's complement methods, Binary arithmetic. Boolean algebra: De Morgan's Theorem, Various logic gates and their truth tables and circuits, Representation in SOP and POS forms, Canonical forms, Minimization of logic expressions by algebraic method, Karnaugh maps up to 6 variables.

MODULE 2: Combinational logic circuits (8 Lectures)

Combinational logic circuit design using truth-table, Different Adders and Subtractors, Comparator, Encoder, Decoder, Multiplexer, Demultiplexer, Barrel shifter and ALU, Driver & Multiplexed Display.

MODULE 3: Sequential logic circuits (20 Lectures)

Building blocks like S-R, JK and Master-Slave JK FF, Edge triggered FF, Ripple and Synchronous counters, shift registers, Finite state machines, Design of synchronous FSM, Algorithmic State Machines charts, Designing synchronous circuits like Pulse train generator, Pseudo Random Binary Sequence generator.

MODULE 4: Logic Families and Semiconductor Memories (7 Lectures)

Brief idea about DTL, TTL, ECL, MOS and CMOS families and their comparison based on Parameters: fan-in, fan-out, propagation delay, speed-power product, etc. TTL NAND gate, Tristate TTL, ECL, CMOS families and their interfacing. Memory elements, Concept of Programmable logic devices like FPGA. Logic implementation using Programmable Devices.

Suggested Text/ Reference Books:

1. R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition, 2009.
2. Douglas Perry, "VHDL", Tata McGraw Hill, 4th edition, 2002.
3. W.H. Gothmann, "Digital Electronics-An introduction to theory and practice", PHI, 2nd Edition, 2006.
4. D.V. Hall, "Digital Circuits and Systems", Tata McGraw Hill, 1989
5. Charles Roth, "Digital System Design using VHDL", Tata McGraw Hill 2nd edition 2012.
6. Anand Kumar, "Fundamentals of Digital Logic", PHI.
7. Morris Mano and Michael D Ciletti, "Digital Design", 4th Edition, Pearson, 2008. Zvi Kohavi and Niraj K Jha, "Switching and Automata Theory", Cambridge

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241303	Electrical and Electronics Measurements	3-0-0	3

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** To understand concepts of various electrical and electronic measuring instruments.
- **CO2:** To familiarize with different electromechanical and electronic instruments.
- **CO3:** To introduce instruments for power, energy measurements and special type meters.
- **CO4:** To explain instrument transformers and magnetic measurements.
- **CO5:** To be able to measure different physical parameters with the help of AC bridges.
- **CO6:** To understand construction and working principle of electronic instruments.

MODULE 1: Electrical Standards (2 lectures)

Standards of e.m.f. and resistance, frequency dependence of resistance, Inductance and Capacitance, Time and frequency standards, Qualities of Measurements.

MODULE 2: Electromechanical Indicating Instruments (7 lectures)

PMMC galvanometer, Ohmmeter, Electrodynamometer, Moving iron meter, Rectifier and thermo-instruments, Comparison of various types of indicating instruments.

MODULE 3: Power and Energy Measurement (6 lectures)

Electrodynamometer type of wattmeter and power factor meter, Power in poly phase system: two wattmeter method, Single-phase induction and Electronic energy meters. Special Type meters: Construction and working principle of frequency meter, synchroscope, power factor meter, flux meter, maximum demand meter.

MODULE 4: Instrument Transformers (6 lectures)

Current & Voltage transformers, Constructional features, Ratio & Phase angle errors, errors compensation, testing of CT and PT Magnetic Measurements: Determination of BH curve and Hysteresis loop, Measurement of iron losses with Lloyd Fisher Square

MODULE 5: Bridge Measurements (9 lectures)

AC bridges: Applications and conditions for balance, Maxwell's bridge, Hay's bridge, Schering bridge, Wien's bridge, De Sauty's bridge, Insulation testing, Ground resistance measurement, Varley and Murray loop test.

MODULE 6: Electronic Instruments (10 lectures)

Electronic Voltmeters, Digital voltmeters, Electronic multimeter, general characteristics ramp type voltmeter, quantization error, digital frequency meter/timer, Q meter and its applications, distortion meter, wavemeter and spectrum Analyzer, block diagram and applications of oscilloscopes, vertical and horizontal deflection systems, delay line, multiple trace, CRO probe & transducers, measurement of voltage, current, phase and frequency by CRO, storage oscilloscopes, working principle and circuit diagram of strip-chart and X-Y recorders

Suggested Text/ Reference Books:

1. Golding, E.W., and Widdis, F.C., Electrical Measurements and Measuring Instruments, Pitman (2003).
2. Helfrick, A.D., and Cooper, W.D., Modern Electronic Instrumentation and Measurement Techniques, Prentice Hall of India (2007).
3. D.V.S. Murthy, "Transducers and Instrumentation", Prentice Hall of India, 2nd edition, 2009.
K. Sawhney, "A course in Electrical and Electronic Measurement and Instrumentation", Dhanpatrai & Co., 12th edition, 2010.
4. Kalsi, H.S., Electronic Instrumentation, TMH 2007.
5. D O Doebelin, "Measurements Systems, Applications and Design", TMH Publications, 5th edition, 2003.

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241304	Network Analysis	3-1-0	4

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Apply network theorems in network analysis.
- **CO2:** Apply the frequency domain and time domain techniques for network analysis.
- **CO3:** Determine different network functions.
- **CO4:** Apply the knowledge of coupled circuits, Network Topology in system design.
- **CO5:** Design passive filters.

MODULE 1: An Overview of Network Theorems (10 Lectures)

Nodal and Mesh Analysis, matrix approach of network containing voltage sources, current sources and reactance, source transformation and duality. Network theorems: Superposition, reciprocity, Thevenin's, Norton's, Maximum Power Transfer, Compensation and Tellegen's theorem as applied to AC circuits. Modified Nodal Analysis, three phase circuit analysis: three phase balanced and unbalanced circuit and power calculation.

MODULE 2: Trigonometric and Exponential Fourier series (4 Lectures)

Discrete spectra and symmetry of waveform, steadystate response of a network to non-sinusoidal periodic inputs, power factor, effective values, Fourier transform and continuous spectra.

MODULE 3: Laplace Transform for Network Analysis (5 Lectures)

Complete response (steady state and transient response) of a network from the S-domain circuit, Linear constant co-efficient differential equation and continuous time domain analysis of RC, RL, and RLC networks with and without initial conditions using Laplace transform.

MODULE 4: Two Port Networks and Network Functions (8 Lectures)

General principle; Z, Y, Hybrid, Transmission and image parameters and equivalent circuits; Interrelationship between various parameters; Series, parallel and cascade connection of networks and parameters; Driving point and transfer Impedance and Admittance function; Pole-Zeros of Network function, Determination of time domain behavior from pole-zero plot; Magnitude and phase plot of network functions.

MODULE 5: Resonance and Coupled Circuits (4 Lectures)

Resonance—series and parallel; Q factor, BW; Analytical procedure for solving coupled circuits; Mutual Inductance, coefficient of coupling; Single tuned and double tuned circuits; Effects of over coupling and selectivity curve; Ideal transformer.

MODULE 6: Passive Filter (4 Lectures)

Filter fundamentals, Classification of filters—Low pass, High Pass, Band Pass and Band Reject Filters.

MODULE 7: Network Topology (5 Lectures)

Graph of a network; Concept of tree, co-tree link, chord, forest, co-forest; Planar and non-planar graph; Incidence matrix, tie set matrix, cut set matrix; Fundamental cut set and tie set schedule; Introduction to equation formulation graphically; Proofs of Nodal and Mesh Analysis; Duality of network.

Suggested Text/ Reference Books:

1. Van, Valkenburg.; "Network analysis"; Prentice hall of India, 2000
2. Sudhakar, A., Shyammmohan, S. P.; "Circuits and Network"; Tata McGraw-Hill New Delhi, 1994
3. A William Hayt, "Engineering Circuit Analysis" 8th Edition, McGraw-Hill Education

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305B	Mathematics III-B	3-0-0	3

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Apply concepts of vector spaces, linear transformations, and inner product spaces to analyse linear systems and perform orthogonalization in engineering and scientific computations.
- **CO2:** Apply fundamental probability concepts, including conditional probability, independence, and Bayes' Theorem, to model and analyse random events in engineering contexts.
- **CO3:** Analyse discrete and continuous random variables using probability distributions and joint distributions, and apply these concepts to model uncertainties in real-world systems.
- **CO4:** Compute expectation, variance, moments, and covariance to quantify and interpret the statistical behaviour of random variables in practical scenarios.
- **CO5:** Model and analyse random processes by evaluating stationarity, ergodicity, and correlation functions to address uncertainties in engineering systems.

MODULE 1: Linear Algebra: (12 Hours)

Vector space: Definition and properties of vector spaces, definition and properties of subspace, linear independence, bases and dimension.

Linear transformation: Definition, linear maps and matrices, rank-nullity theorem, Eigen values and Eigen vectors.

Inner product space: Definition, Euclidean and unitary space, Schwartz's inequality, orthogonality, norms, orthonormal bases, Gram-Schmidt orthogonalization process.

MODULE 2: Probability: (5 Hours)

Classical and axiomatic definitions, sample space, joint probability, conditional probability, Bayes' Theorem, independent events.

MODULE 3: Random variables: (12 Hours)

Definition, continuous and discrete random variables, cumulative distribution function (CDF), probability functions (PMF & PDF), Probability Distributions: Gaussian, uniform and exponential, binomial distribution, Poisson, joint distributions (JPMF & JPDF), central limit theorem (Statement only).

MODULE 4: Operation on one random variable: (6 Hours)

Expectation, variance and standard deviation, Covariance, Raw moment & central moments.

MODULE 5: Random Process: (10 Hours)

The random process concept, stationarity, wide-sense stationarity, ergodicity, auto-correlation and cross-correlation functions, auto-covariance and cross-covariance functions, examples of random processes.

Textbooks/References:

1. Introduction to Linear Algebra: Strang G., Wellesley-Cambridge Press; 4th Edition, 2009
2. Linear Algebra and its Application: David C Lay, Pearson.
3. Probability, Random Variables and Random Signal Principles, Peyton Z Peebles Jr. , McGraw-Hill, 4th Edition, 2017.
4. Probability and Random Processes with Applications to Signal Processing: Henry Stark and John W. Woods, Pearson, 3rd Edition, 2012.
5. Advanced Engineering Mathematics: Erwin Kreyszig, John Wiley & Sons, 9th Edition, 2006.
6. Introduction to Probability Theory: P. G. Hoel, S. C. Port and C. J. Stone, Universal Book Stall, 2003 (Reprint).

7. A text book of Engineering Mathematics: N.P. Bali and Manish Goyal, Laxmi Publications, 2010 (Reprint).
8. Engineering Mathematics (for semester III): Veerarajan T., Tata McGraw-Hill, 2010.
9. Introduction to Probability and Statistics for Engineering and Scientists: S. M. Ross, Academic Press, 3rd Edition, 2004.
10. Probability, Random Variables, and Stochastic Processes: A. Papoulis and S. U. Pillai, Tata McGraw-Hill, 4th Edition, 2002.

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text Books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

- NPTEL / SWAYAM courses on IKS
- AICTE-IKS Portal (<https://iksindia.org/>)
- Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text Books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241311	Semiconductor Devices and Circuits Lab	0-0-2	1

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Experiment with different measuring instruments such as diodes, transistors etc.
- **CO2:** Design the circuits with basic semiconductor devices.
- **CO3:** Measure and record the experimental data, analyse the results, and prepare a formal laboratory report.

LIST OF LABORATORY EXPERIMENTS/DEMONSTRATIONS

1. To study the different components and equipment of Electronics Laboratory.
2. To study V_D - I_D characteristics of Silicon Diode.
3. To study V-I Characteristics of Zener diode and Zener diode as a voltage regulator.
4. To study Half-Wave rectifier with and without filter.
5. To study Full-Wave rectifier with and without filter.
6. To study Bridge rectifier with and without filter.
7. To plot the input and output characteristics of a transistor in CB Configuration.
8. To plot the input and output characteristics of a transistor in CE Configuration.
9. Comparison of Performance of Self Bias and Fixed Bias Circuits.
10. To study Drain characteristics and Transfer characteristics of JFET.

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241312	Digital Electronics Lab	0-0-2	1

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Simplify, design and implement Boolean expression/half and full adders using basic/universal gates.
- **CO2:** Design and implement the various combinational circuits using MSI components.
- **CO3:** Design and implement the various sequential circuits.
- **CO4:** Analyze the results, and prepare a formal laboratory report.

LIST OF LABORATORY EXPERIMENTS/DEMONSTRATIONS

1. To familiarize with logic gate IC packages and to verify the truth tables of logic gates.
2. To verify Demorgan's theorem for 2 variables.
3. To design and set up half adder and half Subtractor using a. EXOR gates and gates b. NAND gates.
4. To design and set up Full adder and Full Subtractor using a. EXOR gates and gates b. NAND gates.
5. Design and implementation of parallel adder/Subtractor using IC7483.
6. Design and implementation of a. BCD-to-excess-3code converter and vice versa. b. gray-to-binary and vice-versa.
7. Design and implementation of one bit, two-bit magnitude comparators.
8. Design and set up a Multiplexer (MUX) using gates and ICs.
9. Implementation and verification of truth table for J-K flip-flop, Master-slave J-K flip-flop, D flip-flop and T flip-flop.
10. Design and implementation of Mod-N synchronous counter using J-K flip-flops.
11. Design and implementation of shift register to function as a) SISO, b) SIPO c) PISO d) PIPO e) shift left and f) shift right operation.
12. Design and implementation of a) Ring counter and b) Johnson counter using 4-bit shift register

Course Code	Course Title	Hours per week L-T-P	Credit C
PE241313	Electrical and Electronics Measurements Lab	0-0-2	1

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1:** Measure low resistance, self-inductance and capacitance using different bridges
- **CO2:** Learn calibration and testing of different types of meters.
- **CO3:** Learn about measurement of power and power factor in single and multi-phase system.

LIST OF LABORATORY EXPERIMENTS/DEMONSTRATIONS

1. Measurement of Low Resistance by Kelvin's Double Bridge Method.
2. Measurement of Self Inductance and Capacitance using Bridges.
3. Study of Galvanometer and Determination of Sensitivity and Galvanometer Constants.
4. Calibration of Voltmeters and Ammeters using Potentiometers.
5. Testing of Energy meters (Single phase type).
6. Measurement of Iron Loss from B-H Curve by using CRO.
7. Measurement of R, L, and C using Q-meter.
8. Measurement of Power in a single phase circuit by using CTs and PTs.
9. Measurement of Power and Power Factor in a three phase AC circuit by two-wattmeter method.
10. Study of Spectrum Analyzers.
